THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 67 **(b)** 81 **(c)** 53 **(d)** 109

2. If $x - y = 3$ and $xy = 20$, then $x^2 + y^2$ equals: 1

(a) 49 **(b)** -31 **(c)** 9 **(d)** 29

3. The value of $512^2 - 488^2$ is: 1

(a) 576 **(b)** 500288 **(c)** 24100 **(d)** 24000

4. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 8 **(b)** 7 **(c)** 11 **(d)** 9

5. If $(x + p)(x + q) = x^2 + 7x + 12$, then $p^2 + q^2$ equals: 1

(a) 73 **(b)** 37 **(c)** 25 **(d)** 49

6. The area of a square of side 25 cm exceeds that of a square of side 22 cm by: 1

(a) 484 **(b)** 9 **(c)** 75 **(d)** 141

7. The value of $24^2 - 23^2$ is: 1

(a) 24 **(b)** 47 **(c)** 49 **(d)** 48

8. $(a + b)^2 = a^2 + b^2$ is true only when: 1

(a) $a = b$ **(b)** $ab = 0$ **(c)** $a = 1$ **(d)** $a + b = 0$

9. **Assertion (A):** $(x + 5)^2 = x^2 + 10x + 25$. 1

Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $(a-b)^2 = (b-a)^2$. 1

Reason (R): $(a+b)(a-b) = a^2 - b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x + y = 12$ and $xy = 35$, find $x^2 + y^2$. 2

12. Evaluate $612^2 - 388^2$ using an identity. 2

13. If $(x+p)(x+q) = x^2 + 8x + 15$, find $p^2 + q^2$. 2

14. If $x + y = 8$ and $xy = 15$, find $x^2 + y^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x - y = 4$ and $xy = 21$, find $x^2 + y^2$ and $(x+y)^2$. 3

16. Simplify $(3x+2)^2 - (3x-2)^2$. 3

17. If $a + b = 10$ and $ab = 24$, find $a^2 + b^2$ and $(a-b)^2$. 3

OR

If $x - y = 6$ and $xy = 40$, find $x^2 + y^2$ and $(x+y)^2$. 3

18. Simplify $(4x+5)^2 - (4x-5)^2$. 3

OR

If $a + b = 13$ and $ab = 40$, find $a^2 + b^2$ and $(a-b)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rectangular vegetable plot has length $(x+5)$ m and breadth $(x+2)$ m. A square flower bed beside it has side $(x+3)$ m. 5

(i) Write the area of the vegetable plot as an expression.

(ii) Find its area when $x = 10$.

(iii) Write the perimeter of the plot.

(iv) By how much does the area of the plot exceed that of the flower bed? Verify for $x = 10$.

20. The sides of a rectangle are $(4x+3)$ m and $(4x-3)$ m. Find its area as an expression, its value at $x = 6$, and by how much it is less than the area of a square of side $4x$ m. 5

OR

If $x + y = 16$ and $x^2 + y^2 = 146$, find xy, $(x-y)^2$ and $x - y$ (take $x > y$). 5

END OF QUESTION PAPER — Total: 40 Marks